

Zonal vs. Nodal Pricing: An Analysis of Different Pricing Rules in the German Day-Ahead Market

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Abstract—How do pricing and settlement rules shape price signals and costs in electricity markets with large bidding zones and non-convex generation? This question has gained urgency in Europe as rising congestion motivated the European Commission’s Bidding Zone Review (BZR). Using the publicly released BZR dataset for 2025, we analyze nodal and zonal market designs for the German–Luxembourgish bidding zone.

We compare Integer Programming, Convex Hull, and Join pricing across multiple zonal configurations and against a nodal benchmark, focusing on day-ahead prices, price dispersion, uplifts, production and redispatch costs. We find that alternative zonal splits lead to only modest changes in average prices and dispersion, with limited cost differences between single- and multi-zone designs. In contrast, pricing rules and uplift mechanisms play a central role in shaping price signals in non-convex markets. Our results indicate that European policymakers debating non-uniform pricing should prioritize settlement design alongside bidding zone reconfiguration.

Index Terms—Energy Economics and Policy, Nodal Pricing, Non-Convex Markets, Zonal Pricing

I. INTRODUCTION

Electricity market design determines how network constraints are reflected in prices and settlements. A key choice is between *nodal pricing* (locational marginal pricing, LMP), where electricity is priced at individual network nodes, and *zonal pricing*, where nodes are aggregated into bidding zones with uniform prices. While nodal pricing is used in many wholesale markets (including most U.S. regions), Europe’s day-ahead (DA) market relies on zonal pricing. Zonal designs simplify clearing and support liquidity, but they abstract from intra-zonal congestion, which can render DA dispatch infeasible and shifts congestion management to ex post redispatch conducted by Transmission System Operators (TSOs). In contrast, nodal pricing internalizes network constraints in DA clearing and reduces reliance on redispatch. In U.S. markets, rising congestion and uplift concerns were important drivers behind the transition from zonal to nodal pricing [1].

In Europe, the policy debate is not primarily about a wholesale move to nodal pricing, but about whether alternative bidding-zone configurations and settlement rules can improve congestion management while preserving market integration [2], [3]. This question has become more pressing with the energy transition: high shares of variable renewables and limited transmission expansion increase congestion frequency [4]. We focus on the German–Luxembourgish bidding zone (BZ), where north–south imbalances in supply and demand, as well as the transmission bottlenecks have led to sharply rising redispatch volumes and costs (34 TWh and EUR 3.1 billion in 2023) [5]. These developments motivated the European Commission’s Bidding Zone Review (BZR).¹ A central BZR output is a large-scale LMP study by ENTSO-E that provides nodal prices from a linearized unit commitment model for the target year 2025 [7]. Based on this assessment, ACER proposed alternative German zone splits including 2–4 zones [8] which are displayed in Figure 1², and non-confidential BZR data have been released for independent analysis [7].

A second, closely related dimension concerns *pricing rules* in non-convex markets. Unit commitment non-convexities complicate settlement and can require make-whole payments or distort congestion signals. Established approaches such as Integer Programming (IP) and Convex Hull (CH) pricing address these issues with different trade-offs [9], [10], and are currently discussed in Europe in the context of non-uniform pricing [11]. Yet, empirical evidence on how pricing rules perform comparably across nodal and zonal designs on realistic European-scale data remains limited. Existing studies on German BZ splits often rely on proprietary models and report north–south price spreads of roughly 5–15 EUR/MWh under heterogeneous assumptions [12]–[14], while system-level congestion and redispatch studies typically abstract from settlement rules [15], [16]. Conversely, much of the pricing-

¹The final BZR results were released in April 2025 [6].

²The alternative zonal configurations were obtained by solving a clustering problem with time series of nodal prices as features.

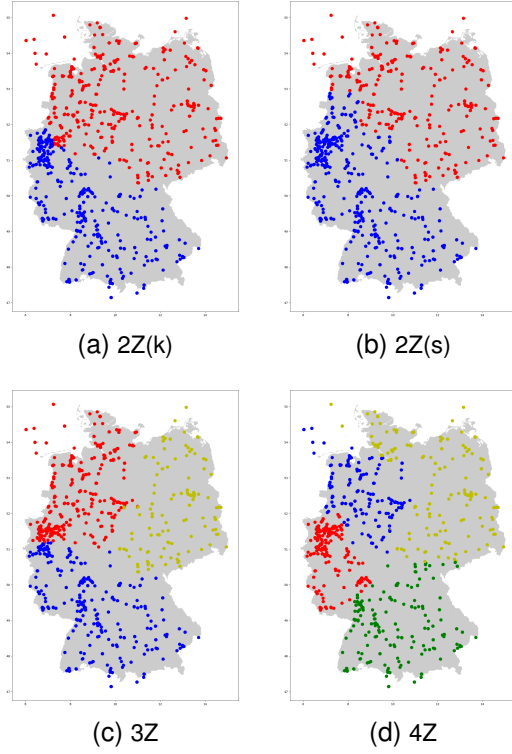


Fig. 1. Zonal configurations.

rule literature is grounded in U.S. nodal settings [17].

This paper provides a controlled, data-driven comparison using the public BZR dataset. Focusing on Germany, we apply multiple pricing rules across nodal and zonal market designs and across the BZR-relevant two- to four-zone configurations. Our contributions are threefold: (i) we distinguish dispatch outcomes from pricing/settlement outcomes under a common dispatch approximation; (ii) we evaluate IP, CH, and Join pricing across both nodal and zonal configurations; and (iii) we quantify how zoning and pricing rules affect price dispersion, uplift requirements, and day-ahead cost components on a large-scale, policy-relevant dataset. Our analysis is not a comprehensive welfare evaluation as balancing, and long-run investment responses are not modeled explicitly. Instead, the paper provides a transparent stress test of pricing-rule behavior under realistic data to inform ongoing debates on bidding-zone design and settlement reform.

II. METHODOLOGY

A. Day-Ahead Market Clearing Models

We model the day-ahead market as a unit commitment problem with inelastic demand and a DC power flow approximation (DCOPF) [18]. The market outcome is determined by minimizing total generation costs (or, maximizing the welfare) subject to network feasibility and unit commitment constraints.

1) *Nodal market clearing*: Nodal clearing represents power balance at each node and time and enforces transmission limits on all lines. Commitment variables u_{st} and dispatch $y_{s,nst}$ are linked through capacity constraints, with minimum up-time

captured via start-up indicators ϕ_{st} . The resulting formulation is a mixed-integer linear program (MILP):

$$\min_{x,y,u,\theta} \sum_{s \in S} \sum_{t \in T} g_s y_{s,nst} + \sum_{s \in S} \sum_{t \in T} h_s u_{st} \quad (\text{Nodal Clearing})$$

$$\text{s.t.} \quad \sum_{s \in S} y_{s,nt} - \sum_{b \in B} x_{b,nt} = \sum_{m \in N} -B_{nm}(\theta_{nt} - \theta_{mt}), \quad \forall n, t \quad (1)$$

$$\underline{F}_{nm} \leq B_{nm}(\theta_{nt} - \theta_{mt}) \leq \overline{F}_{nm}, \quad \forall n, m, t \quad (2)$$

$$\underline{P}_{st} u_{st} \leq y_{s,nst} \leq \overline{P}_{st} u_{st}, \quad \forall s, t \quad (3)$$

$$y_{s,nt} = 0, \quad \forall s, n \neq n_s, t \quad (4)$$

$$\phi_{st} \geq u_{st} - u_{s(t-1)}, \quad \forall s, t \neq T_0 \quad (5)$$

$$\sum_{i=t-\underline{R}_s+1}^t \phi_{si} \leq u_{st}, \quad \forall s, t \neq T_0 \quad (6)$$

$$x_{b,nbt} = P_{bt}, \quad \forall b, t \quad (7)$$

$$x_{b,nt} = 0, \quad \forall b, n \neq n_b, t \quad (8)$$

$$\theta_{n^*t} = 0 \quad \forall t \quad (9)$$

To proxy operational security requirements, we derate all thermal limits by 20% (i.e., we enforce a uniform security margin). A DCOPF-feasible solution may still be ACOPF-infeasible and require further adjustments to the allocation, which we disregard in this work.

2) *Zonal market clearing*: To represent zonal pricing, we aggregate nodes into zones and enforce network feasibility only on inter-zonal lines. Intra-zonal constraints are not binding in the DA clearing and feasibility is instead restored ex post via redispatch. We treat all interconnectors between zones as critical network elements and apply a uniform derating of 20% to their capacities, in line with stylized capacity-availability assumptions from [14], [19]. The zonal clearing problem is:

$$\min_{x,y,u,\theta} \sum_{s \in S} \sum_{t \in T} g_s y_{s,nst} + \sum_{s \in S} \sum_{t \in T} h_s u_{st} \quad (\text{Zonal Clearing})$$

$$\text{s.t.} \quad \sum_{s \in S} y_{s,zt} - \sum_{b \in B} x_{b,zt} = \sum_{\substack{n \in z \\ m \notin z}} -B_{nm}(\theta_{nt} - \theta_{mt}), \quad \forall z, t \quad (10)$$

$$\underline{F}_{nm} \leq B_{nm}(\theta_{nt} - \theta_{mt}) \leq \overline{F}_{nm}, \quad \forall n, m : z(n) \neq z(m), t \quad (11)$$

$$\theta_{z^*t} = 0, \quad \forall t \quad (12)$$

$$(3) - (8).$$

3) *Redispatch*: Zonal DA outcomes are generally not physically feasible under full network constraints and require redispatch. In Germany, redispatch is implemented as a cost-based mechanism. While CACM Regulation Article 35 and §13 EnWG require redispatch measures to minimize overall costs, neither specifies the operational optimization problem, and the detailed TSO procedures are not publicly disclosed. The literature considers various redispatch models [20]–[22], which differ in the set of redispatchable generators, the penalties associated with deviations from the zonal dispatch, and the

treatment of RES curtailment. To obtain a benchmark for comparing zonal and nodal outcomes, we implement a minimum-cost, unconstrained redispatch model following [20]. Starting from the zonal dispatch (y^Z, u^Z) , we re-optimize generation subject to full nodal network constraints:

$$\min_{x, y, u, \theta} \sum_{s \in S} \left(\sum_{t \in T} g_s(y_{s, n_{st}} - y_{s, n_{st}}^Z) + \sum_{t \in T} h_s(u_{st} - u_{st}^Z) \right) \quad (13)$$

s.t. (1) – (9).

All generators are redispatchable. Since we abstract from institutional and technical restrictions, such as limited redispatchable units and curtailment constraints, the model provides an optimistic lower bound on redispatch costs rather than a detailed operational representation.

B. Pricing and Settlement in Non-Convex Markets

In markets with unit commitment non-convexities, linear prices typically fail to support the welfare-maximizing allocation as a Walrasian equilibrium³ [23]. Consequently, market operators adopt pricing rules that approximate such an equilibrium by balancing three incentive and settlement metrics, which are defined for a fixed allocation or dispatch (x, y, u, θ) and prices p [24]:

- **Make-Whole Payments (MWP):** Out-of-market payments required to compensate paradoxically accepted generators whose production costs exceed market revenues. High MWPs increase the uplift charges levied on consumers.
- **Global Lost Opportunity Costs (GLOCs):** The profit a generator foregoes by following the dispatch instead of their own profit-maximizing schedule at market prices p . High GLOCs signal a lack of envy-freeness.
- **Local Lost Opportunity Costs (LLOCs):** A subset of GLOCs representing the profit forgone given the current commitment status u . Non-zero LLOCs indicate that prices do not accurately reflect the marginal value of transmission, thereby distorting congestion signals.

We evaluate three established pricing rules that navigate these trade-offs differently. *Integer Programming (IP) pricing* [9] sets prices by fixing optimal commitment decisions and solving the resulting linearized dual. While IP ensures zero LLOCs (preserving sharp congestion signals), it often results in high MWPs. *Convex Hull (CH) pricing* [10], [25] minimizes total GLOCs by pricing the convex envelope of the cost function, but it typically incurs non-zero LLOCs. Finally, *Join pricing* [24] utilizes a multi-objective dual formulation to minimize both MWPs and LLOCs simultaneously. Our pricing implementation is based on [24].

For comparison with the European design, we implement an *Euphemia-style procedure* [26]. Unlike the previous pricing algorithms, Euphemia avoids MWPs by iteratively modifying the allocation to disable paradoxically accepted generators.

³A Walrasian equilibrium is a price-allocation pair in which markets clear, all participants maximize their utility at the prevailing prices (envy-freeness), and payments balance between buyers and sellers.

This ensures a no-uplift settlement but introduces an *allocative loss* relative to the welfare-maximizing dispatch. In our implementation, we first solve the zonal clearing problem, compute IP prices, and check whether any accepted generator has negative utility. If so, we add a cut setting the corresponding commitment variable to zero and re-solve the zonal problem. This cycle continues until no paradoxically accepted generator remains.

III. RESULTS

We use the BZR dataset and focus on the German–Luxembourgish BZ [27], which includes 1687 nodes and 3232 lines. We solve the models in Python 3.11 using Gurobi and simulate all eight weeks of hourly operation available in the BZR scenario year 2009. To maintain computational feasibility for MILP solves, we allow a 5% MIP gap. For summary statistics, we cap reported prices at 100 EUR/MWh to limit the influence of extreme outliers on averages and standard deviations, consistent with BZR reporting practice.

A. Price Levels and Price Dispersion

Table I reports average day-ahead prices and standard deviation across all simulated hours. For reference, the mean BZR price for the weeks under consideration is 40.17 EUR/MWh with a standard deviation of 18.51 EUR/MWh, computed bi-hourly based on a linear relaxation in the BZR report.

Pricing		Nat	2Z(k)	2Z(s)	3Z	4Z	Nodal
IP	Mean	45.92	45.45	46.56	44.90	44.37	46.63
	Std	19.76	19.55	19.79	19.45	19.57	19.54
CH	Mean	40.13	40.53	40.81	40.21	41.35	44.79
	Std	14.43	14.42	14.74	14.39	14.32	16.78
Join	Mean	46.31	48.04	46.00	45.51	48.41	47.65
	Std	19.97	20.45	20.12	20.34	20.64	20.06
Eup	Mean	47.08					
	Std	20.61					

TABLE I
AVERAGE PRICES AND STANDARD DEVIATIONS (EUR/MWh).

Across all allocation rules, CH pricing yields slightly lower mean prices and lower standard deviation than IP or Join pricing. This aligns with the fact that CH prices are derived from a convexified representation of generator non-convexities and can be set by relatively cheap units that are not necessarily committed in the integer allocation [28]. IP and Join prices exhibit comparable average levels.

B. Zonal configurations and price separation

Price distributions are broadly similar across the zonal configurations for a fixed pricing rule. In many hours, inter-zonal constraints are not binding, implying that zonal prices coincide with the national (single-zone) price. However, average statistics can mask stressed hours in which price separation becomes substantial. To make tail behavior explicit, Table II reports mean and standard deviation of IP prices per zone.

A key driver of limited average price separation is the assumed availability of inter-zonal transfer capacity in the zonal clearing model. In our implementation, we model interconnectors individually and approximate security margins via a uniform 20% derating. Accordingly, the frequency and

Zone		Nat	2Z(k)	2Z(s)	3Z	4Z
Zone 1	Mean	45.92	46.88	48.36	45.42	42.43
	Std	19.76	19.69	19.87	19.68	19.22
Zone 2	Mean		44.02	44.76	44.28	45.58
	Std		19.30	19.55	19.37	19.96
Zone 3	Mean				45.02	44.08
	Std				19.28	19.05
Zone 4	Mean					45.38
	Std					19.86

TABLE II
IP ZONAL PRICES: MEAN AND STANDARD DEVIATION PER ZONE (EUR/MWh).

magnitude of price separation are conditional on this approximation and on the modeled set of critical network elements. Consistent with this interpretation, sensitivity checks in which inter-zonal transfer is restricted more strongly produce substantially larger north–south price spreads in stressed hours (sometimes exceeding 30 EUR/MWh), while leaving average prices broadly comparable.

C. Nodal price patterns

Under nodal pricing, prices vary across locations within each hour, which creates locational incentives for flexible generation, demand response, and storage [29]. The share of capped outliers is higher under nodal pricing (e.g., 0.25% of IP prices capped at 100 EUR/MWh) than under zonal designs (0.13% for IP), reflecting tighter local constraints and the larger number of priced locations. Figures 2 map average prices geographically. Average nodal prices tend to be higher in southern Germany, consistent with a north–south difference in RES generation and congestion.

D. Variance decomposition: congestion vs. time

A primary goal of bidding-zone design is to reduce price variance within zones [30]. Price variance can be driven both by (i) spatial congestion in a given hour and (ii) temporal variation in supply and demand at a location. Figure 3 contrasts these two components for IP pricing by showing the distribution of hourly standard deviations of nodal prices across nodes (congestion-based dispersion) and of time-series standard deviations at each node (time-based dispersion).

Time-based dispersion exceeds congestion-based dispersion: for IP prices, the mean congestion-based standard deviation is 12.68 EUR/MWh, compared to a mean time-based standard deviation of 17.39 EUR/MWh. This indicates that temporal demand–supply fluctuations contribute more to overall price variance than spatial congestion under the adopted dispatch approximation.

Importantly, day-ahead prices alone do not determine the desirability of a market design or bidding-zone configuration [31]. They do not reflect redispatch and balancing costs, network tariffs used to fund system operations, congestion rent allocation, or the settlement adjustments required to address non-convexities. Accordingly, the results above are best interpreted as evidence on how alternative pricing and settlement rules shape price signals and dispersion under a realistic dataset and transparent dispatch approximations.

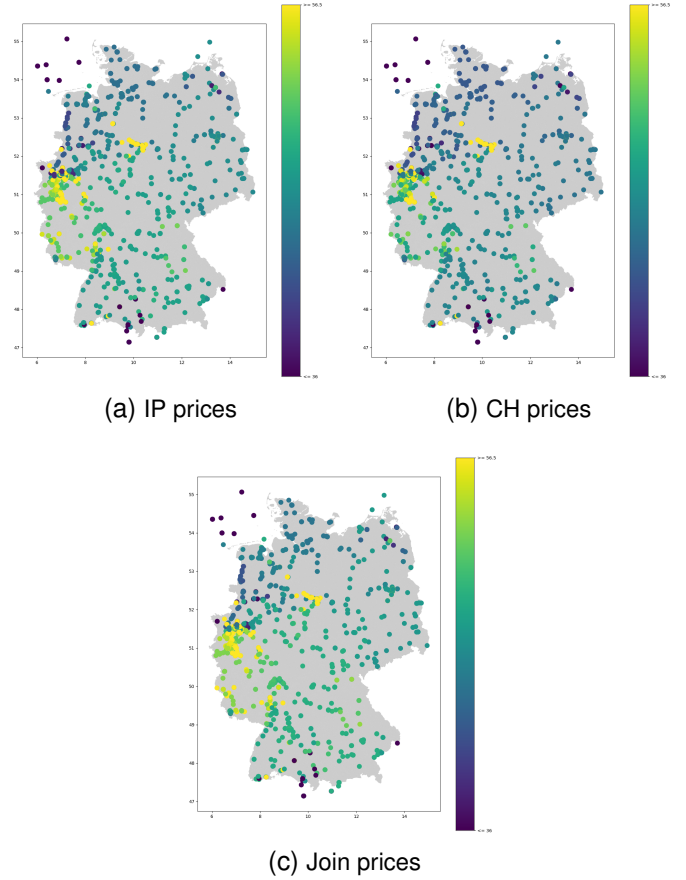


Fig. 2. Mean nodal prices [EUR/MWh].

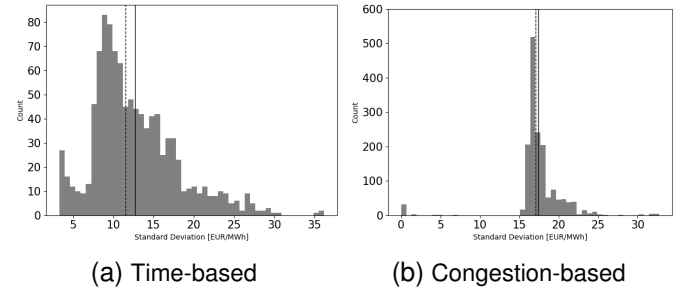


Fig. 3. Histogram of hourly standard deviations for IP prices across hours (time-based standard deviation) and nodes (congestion-based standard deviation) in EUR/MWh.

E. Uplift and Incentive Metrics: GLOCs, LLOCs, and MWPs

Table III summarizes average daily GLOCs, LLOCs, and MWPs. By construction, CH pricing minimizes total GLOCs, and the results confirm that CH yields the lowest GLOC levels. However, CH also exhibits the highest LLOCs, consistent with prior observations that CH prices can weaken congestion signals by allowing prices to be influenced by units that are not committed in the integer allocation [28]. In contrast, IP pricing yields zero LLOCs, meaning that, under fixed commitment, generators have no profitable local deviation and nodal price differences across adjacent nodes arise only when the connecting line is congested. Join pricing substantially

		Nat	2Z(k)	2Z(s)	3Z	4Z	Nodal
GLOCs	IP	2975.07	2666.81	3332.45	2563.87	3130.78	5469.46
	CH	124.24	162.90	520.53	141.05	717.98	382.10
	Join	981.50	1880.94	1001.44	1801.13	2616.04	4792.71
	Eup	1002.36					
LLOCs	IP	0.00	0.00	0.00	0.00	0.00	0.00
	CH	11.39	31.28	109.17	22.23	174.69	184.63
	Join	0.88	3.54	1.43	1.47	15.49	36.93
	Eup	0.00					
MWPs	IP	32.40	45.45	30.88	21.12	42.16	202.11
	CH	8.69	38.03	78.81	19.79	161.95	114.67
	Join	1.15	7.38	1.78	0.80	1.36	22.11
	Eup	0.00					

TABLE III
AVERAGE DAILY GLOCs, LLOCs, AND MWPs (IN KEUR).

reduces LLOCs compared to CH pricing, indicating improved congestion-signal fidelity, while keeping MWPs low.

Also, Euphemia differs from IP/CH/Join in that it co-determines allocation and prices by iteratively excluding outcomes that involve paradoxical acceptance. In our simplified implementation, this leads to an allocative loss relative to the welfare-maximizing allocation under linear prices within the modeled framework. Quantitatively, the average daily allocative loss of Euphemia relative to the national welfare-maximizing allocation in our samples is 0.34%.

F. Day-Ahead Production and Redispatch Costs

Table IV reports average daily day-ahead production costs (DA) and redispatch costs (RD) under each allocation rule.

	Nat	2Z(k)	2Z(s)	3Z	4Z	Nodal
DA	32,145.08	32,428.55	32,219.17	32,198.95	32,382.74	36,150.70
RD	4,005.62	3,722.15	3,931.53	3,951.75	3,767.96	0

TABLE IV
AVERAGE DAILY DAY-AHEAD PRODUCTION (DA) AND REDISPATCH (RD) COSTS (IN KEUR).

Introducing transmission constraints cannot reduce the cost-minimizing objective. Accordingly, the national model – abstracting from network constraints – yields the lowest day-ahead costs. Adding inter-zonal constraints increases costs only modestly on average for the proposed zonal splits, consistent with the limited frequency of binding inter-zonal constraints under the assumed dispatch approximation. The nodal model internalizes all modeled transmission constraints and therefore results in higher day-ahead production costs.

Redispatch costs are obtained from an unconstrained redispatch model, and thus represent a lower bound on actual redispatch requirements, as the model abstracts from institutional and technical constraints such as limited redispatchable units and curtailment restrictions. Under this optimistic benchmark, zonal configurations exhibit substantial redispatch, whereas nodal clearing eliminates redispatch by construction.

Neither day-ahead production costs nor redispatch costs in isolation constitute welfare measures. Realistic redispatch outcomes depend on TSO-specific operational rules, and a comprehensive total-cost comparison would require detailed institutional modeling beyond the scope of this study. Accordingly, Table IV is interpreted as reporting stylized cost components under a common dispatch approximation, while our primary conclusions concern how pricing and settlement rules shape price signals and uplift requirements.

IV. DISCUSSION

a) *Zonal vs. nodal designs*: Nodal pricing internalizes network constraints in day-ahead clearing and yields slightly higher average prices, while zonal designs abstract from intra-zonal congestion and therefore rely on redispatch to restore feasibility. Across the ACER 2–4 zone configurations, average prices, dispersion, and cost components change only modestly relative to the single-zone benchmark: under the assumed transfer-capacity approximation, inter-zonal constraints are often non-binding and substantial congestion remains within zones. This suggests that moderate zoning reforms are insufficient to align boundaries with structural bottlenecks, consistent with prior evidence [32], [33]. More granular or dynamic zoning concepts could in principle increase price separation [33], [34], but would raise liquidity concerns and require substantial operational changes.

b) *Pricing and settlement in non-convex markets*: Pricing rules affect uplift requirements and congestion signals more strongly than moderate zoning changes. Euphemia avoids MWPs through allocation adjustments but induces allocative losses. CH, IP, and Join pricing trade off GLOCs, LLOCs, and MWPs in different ways, highlighting the relevance of non-uniform pricing in current European debates [11]. In particular, CH minimizes GLOCs but weakens congestion signals, IP preserves congestion signals at the cost of higher MWPs, and Join offers a compromise with low MWPs and small local incentive distortions.

c) *Implications and limitations*: Beyond short-run efficiency, bidding-zone reforms affect regional competitiveness, renewable revenues, and forward-market liquidity [35], [36], and may shape long-run investment in generation, flexibility, and grids. These effects are not captured in our static framework. A full welfare comparison would require realistic redispatch modeling and institutional assumptions that are not publicly available.

V. CONCLUSION

Using the BZR dataset for Germany, we show that proposed zonal splits lead to only modest changes in prices and costs, whereas pricing and settlement rules have more systematic effects on uplift and congestion signals. Our findings underscore that debates on bidding-zone configurations cannot be separated from the choice of pricing rules: from a regulatory perspective, settlement design may have more immediate and predictable impacts than moderate changes in zoning granularity. More broadly, our results suggest that reforms of European day-ahead markets should jointly consider zoning and pricing frameworks rather than treating them as independent design choices.

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REFERENCES

- [1] I. Aravena, Q. L  t  , A. Papavasiliou, and Y. Smeers, "Transmission capacity allocation in zonal electricity markets," *Operations Research*, vol. 69, pp. 1240–1255, 07 2021.
- [2] F. Kalantzis and N. T. Milonas, "Market integration and price dispersion in the european electricity market," in *2010 12th International Conference on the European Energy Market*, 2010, pp. 1–6.
- [3] G. S  nchez-Gonz  lez, A. Bello, J. Reneses, and A. Delgadillo, "Market equilibrium in the european electricity market. a methodology to reduce the network configurations," in *2015 12th International Conference on the European Energy Market (EEM)*, 2015, pp. 1–4.
- [4] K. Neuhoff, J. Barquin, J. W. Bialek, R. Boyd, C. J. Dent, F. Echavarren, T. Grau, C. von Hirschhausen, B. F. Hobbs, F. Kunz, C. Nabe, G. Papaefthymiou, C. Weber, and H. Weigt, "Renewable electric energy integration: Quantifying the value of design of markets for international transmission capacity," *Energy Economics*, vol. 40, pp. 760–772, 2013.
- [5] Bundesnetzagentur, "Bundesnetzagentur ver  ffentlicht daten zum strommarkt 2023," 2024. [Online]. Available: https://www.bundesnetzagentur.de/SharedDocs/Pressemitteilungen/DE/2024/20240103_SMARD.html
- [6] ENTSO-E, "Main report, bidding zone review of the 2025 target year," 2025. [Online]. Available: https://eepublicdownloads.blob.core.windows.net/public-cdn-container/clean-documents/Network%20codes%20documents/NC%20CACM/BZR/2025/Bidding_Zone_Review_of_the_2025_Target_Year.pdf
- [7] —, "Report on the locational marginal pricing study of the bidding zone review process," Brussels, 2022. [Online]. Available: https://eepublicdownloads.blob.core.windows.net/public-cdn-container/clean-documents/Publications/Market%20Committee%20publications/ENTSO-E%20LMP%20Report_publication.pdf
- [8] ACER, "List of alternative bidding zone configurations to be considered for the bidding zone review," 2022. [Online]. Available: https://www.acer.europa.eu/sites/default/files/documents/Individual%20Decisions_annex/ACER%20Decision%2011-2022%20on%20alternative%20BZ%20configurations%20-%20Annex%20I.pdf
- [9] R. P. O'Neill, P. M. Sotkiewicz, B. F. Hobbs, M. H. Rothkopf, and W. R. Stewart, "Efficient market-clearing prices in markets with nonconvexities," *European Journal of Operational Research*, vol. 1, no. 164, pp. 269–285, 2005.
- [10] P. R. Gribik, W. W. Hogan, S. L. Pope *et al.*, "Market-clearing electricity prices and energy uplift," *Cambridge, MA*, 2007.
- [11] All NEMO Committee, "Non-uniform pricing: Explanatory note," 2023. [Online]. Available: <https://www.nemo-committee.eu/assets/files/sdac-non-uniform-pricing-explanatory-note.pdf>
- [12] Aurora, "Power Market Impact of Splitting the German Bidding Zone," 2023. [Online]. Available: <https://auroraer.com/insight/power-market-impact-of-splitting-the-german-bidding-zone/>
- [13] H. Diers, P. Emelianova, P. A. Kienscherf, M. Walde, and J. Zinke, "Price impact of a German bidding zone split," 2023. [Online]. Available: <https://www.ewi.uni-koeln.de/en/publications/price-impact-of-a-german-bidding-zone-split/>
- [14] S. Tiedemann, C. Stiewe, C. Kratzke, L. Hirth, M. Jentsch, N. Damm, N. Gerhardt, and C. Pape, "Gebotszonenteilung: Auswirkungen auf den marktwert der erneuerbaren energien im jahr," 2024.
- [15] G. Thomassen, F. A. C. R. P. C. D., and V. S., "Redispatch and congestion management," Luxembourg (Luxembourg), 2024.
- [16] J. H  rsch, F. Hofmann, D. Schlachtberger, and T. Brown, "Pypsa-eur: An open optimisation model of the european transmission system," *Energy strategy reviews*, vol. 22, pp. 207–215, 2018.
- [17] G. Liberopoulos and P. Andrianesis, "Critical review of pricing schemes in markets with non-convex costs," *Operations Research*, vol. 64, no. 1, pp. 17–31, 2016.
- [18] F. Li and R. Bo, "Dcopf-based lmp simulation: algorithm, comparison with acopf, and sensitivity," *IEEE Transactions on Power Systems*, vol. 22, no. 4, pp. 1475–1485, 2007.
- [19] L. Wyrwoll, K. Kollenda, C. M  ller, and A. Schnettler, "Impact of flow-based market coupling parameters on european electricity markets," in *2018 53rd International Universities Power Engineering Conference (UPEC)*, 2018, pp. 1–6.
- [20] M. Ambrosius, J. Egerer, V. Grimm, and A. H. van der Weijde, "Risk aversion in multilevel electricity market models with different congestion pricing regimes," *Energy Economics*, vol. 105, p. 105701, 2022.
- [21] F. Kunz and A. Zerrahn, "Benefits of coordinating congestion management in electricity transmission networks: Theory and application to germany," *Utilities Policy*, vol. 37, pp. 34–45, 2015.
- [22] C. Byers and G. Hug, "Modeling flow-based market coupling: Base case, redispatch, and unit commitment matter," in *2020 17th International Conference on the European Energy Market (EEM)*, 2020, pp. 1–6.
- [23] K. J. Arrow and G. Debreu, "Existence of an equilibrium for a competitive economy," *Econometrica: Journal of the Econometric Society*, pp. 265–290, 1954.
- [24] M. S. Ahunbay, M. Bichler, and J. Kn  rr, "Pricing optimal outcomes in coupled and non-convex markets: Theory and applications to electricity markets," *Operations Research*, vol. 73, no. 1, pp. 178–193, 2025.
- [25] W. W. Hogan and B. J. Ring, "On minimum-uplift pricing for electricity markets," *Electricity Policy Group*, 2003.
- [26] NEMO Committee, "EUPHEMIA public description," 2024. [Online]. Available: <https://www.nemo-committee.eu/assets/files/euphemia-public-description.pdf>
- [27] ACER, "Annex II CE BZRR Input Data," https://eepublicdownloads.blob.core.windows.net/public-cdn-container/clean-documents/Network%20codes%20documents/NC%20CACM/BZR/2025/Annex_2_CE_BZRR_input_data.pdf, 2025, accessed: 2025-05-07.
- [28] D. A. Schiro, T. Zheng, F. Zhao, and E. Litvinov, "Convex hull pricing in electricity markets: Formulation, analysis, and implementation challenges," *IEEE Transactions on Power Systems*, vol. 31, no. 5, pp. 4068–4075, 2016.
- [29] A. Goyal and K. Bhattacharya, "Design of multi-settlement electricity markets considering demand response and battery energy storage systems participation," *IEEE Transactions on Energy Markets, Policy and Regulation*, vol. 2, no. 2, pp. 226–239, 2024.
- [30] ACER, "ACER's Decision on the alternative bidding zone configurations to be considered in the bidding zone review process. Annex IV. Description of the clustering algorithms," 2022, accessed: 2024-02-09.
- [31] N. Mazzi, A. Lorenzoni, S. Rech, and A. Lazzaretto, "Electricity auctions: A european view on markets and practices," in *2015 12th International Conference on the European Energy Market (EEM)*, 2015, pp. 1–5.
- [32] T. Dobos, M. Bichler, and J. Kn  rr, "Challenges in finding stable price zones in european electricity markets: Aiming to square the circle?" *Applied Energy*, vol. 382, p. 125315, 2025.
- [33] N. Gerhardt, K. Knorr, Y. Harms, J. K. und Diana Bottger, P. Godron, T. Lenck, K. Hartz, F. Huneke, and K. Schaber, "Lokale Strompreise," 2025. [Online]. Available: <https://www.agora-energiawende.de/publikationen/lokale-strompreise>
- [34] S. Risanger, S. J. Bakker, S.-E. Fleten, and A. Tomasgard, "Flexible electricity market clearing zones," *Computers & Operations Research*, vol. 174, p. 106880, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0305054824003526>
- [35] C. Growitsch, S. Nick, and S. Pokorny, "Efficient zonal configurations for european electricity markets. evidence from an empirical evaluation," *Energy Economics*, vol. 44, pp. 387–395, 2014.
- [36] L. Hirth and I. Schlecht, "Market design for variable renewables: Taking storage and flexibility into account," *Energy Economics*, vol. 80, pp. 524–538, 2019.

APPENDIX A NOMENCLATURE

Indices and Sets

$s \in S$	Generator (seller) index and set
$b \in B$	Buyer index and set
$n, m \in N$	Network node (bus) indices and set
$l \in L$	Network line (branch) index and set
$z \in Z$	Zone index and set
$t \in T$	Time period (hour) index and set
$T_0 \in T$	First time period (hour)

Parameters

g_s	Variable cost of seller s in EUR/MWh
h_s	No-load cost of seller s in EUR
n_s, n_b	Nodes at which seller s and buyer b are located
$\underline{R}_s$	Minimum up-time of seller s in hours
$\underline{P}_{st}, \bar{P}_{st}$	Minimum and maximum output of seller s in period t in MW

P_{bt}	Inelastic demand of buyer b in period t in MW
B_{nm}	Susceptance of transmission line (n, m) in pu
$\underline{F}_{nm}, \bar{F}_{nm}$	Minimum and maximum power flow on line (n, m) in MW
n^*, z^*	Reference node and reference zone
$z(n)$	Zone in which node n is located

Decision Variables

$y_{s,nt}$	Power output of seller s at node n in period t in MW
u_{st}	Commitment status of seller s in period t
ϕ_{st}	Start-up indicator of seller s in period t
$x_{b,nt}$	Electricity consumption of buyer b at node n in period t in MW
θ_{nt}	Voltage angle at node n in period t
p_{nt}	Electricity price at node n in period t in EUR/MWh